Chapter 5

Expanding Copilot's Capabilities with Plugins

IN THIS CHAPTER

- **Learning the benefits of using plugins**
- **Finding and installing plugins**
- **Creating your own plugins with Copilot Studio**
- **Seeing some popular Copilot plugins**

In Chapter 4, you learn about creating custom copilots, called *agents*, using Copilot Studio. Custom agents are the way to go when you're building a product that uses Copilot and you want to fully customize it, or when you want anyone to be able to use your agent, regardless of whether they have a Microsoft Copilot 365 license.

The other option for giving Copilot new capabilities is to use (or even create!) plugins. In this chapter, you learn about what plugins are, find out how to create them, and see many different examples of them.

Using Plugins Wisely

Plugins are programs that can be added to a Copilot chat window to give Copilot new capabilities. The main difference between plugins and custom agents is that plugins work within (or "on top of") Microsoft 365

Copilot. Agents, on the other hand, are independent from Microsoft 365 Copilot.

Copilot can use plugins when it deems it necessary or beneficial. Just as a carpenter who has a variety of specialized tools and knows how to use them can do more types of work, Copilot with plugins can handle different types of problems than just Copilot by itself.

In fact, you've already seen some examples of plugins in Copilot. For example, the Web Content plugin gives Copilot the ability to access results from Microsoft Bing, and the Kayak plugin gives Copilot the ability to find flight information and link to the Kayak results.

Understanding orchestration

On a more technical level, a big difference between Copilot plugins and custom agents is that plugins use Copilot's orchestration layer.



The *orchestration layer* is the AI decision-maker that intercepts every prompt you submit to Copilot and decides on the best plugin to handle it. For example, you might submit a request such as the following to Copilot:

> *Use the web to find more information about how to build Copilot plugins.*

Copilot doesn't have a specific rule built into it that looks for every phrase a person might submit that is a request to search the web. Instead, Copilot's orchestration layer uses its understanding of the meaning of the prompt to decide that using the Web Content plugin is the best way to handle the request.

When you create a custom agent, as you can learn about in Chapter 4 of this book, you must specify a trigger that will cause one of the custom or system actions to run.

Just as my dog looks at me with a confused face when I ask him if he'd like to go for a stroll, rather than if he wants to go for a walk, a Copilot agent that's only triggered by the phrase "check my balance" won't react to the phrase "How much money do I have?".



REMEMBER Although it's not enabled by default, you can enable Copilot's AI orchestration within custom agents by enabling Generative AI for an agent. You can learn more about enabling Generative AI in a custom agent in Chapter 4.

Knowing the limitations of LLMs

Large language models (LLMs), such as the one Copilot uses, are fantastic at understanding language and responding. This is what they're made to do. What they're less good at, however, is answering questions that require up-to-date information.

The reason for this is that every LLM has a *knowledge cutoff date*. A knowledge cut-off date is the most recent creation date of the information that was used to train that LLM. Because it takes a long time (and a lot of effort and energy) to train an LLM, the cutoff date may be months or even years behind the current date.

By itself, Copilot doesn't know anything about anything that's happened since its knowledge cutoff date. You can determine the current knowledge cutoff date simply by asking Copilot.

At the time of this writing, when I asked Copilot for its knowledge cutoff date, it told me October 2023, as shown in Figure 5-1.

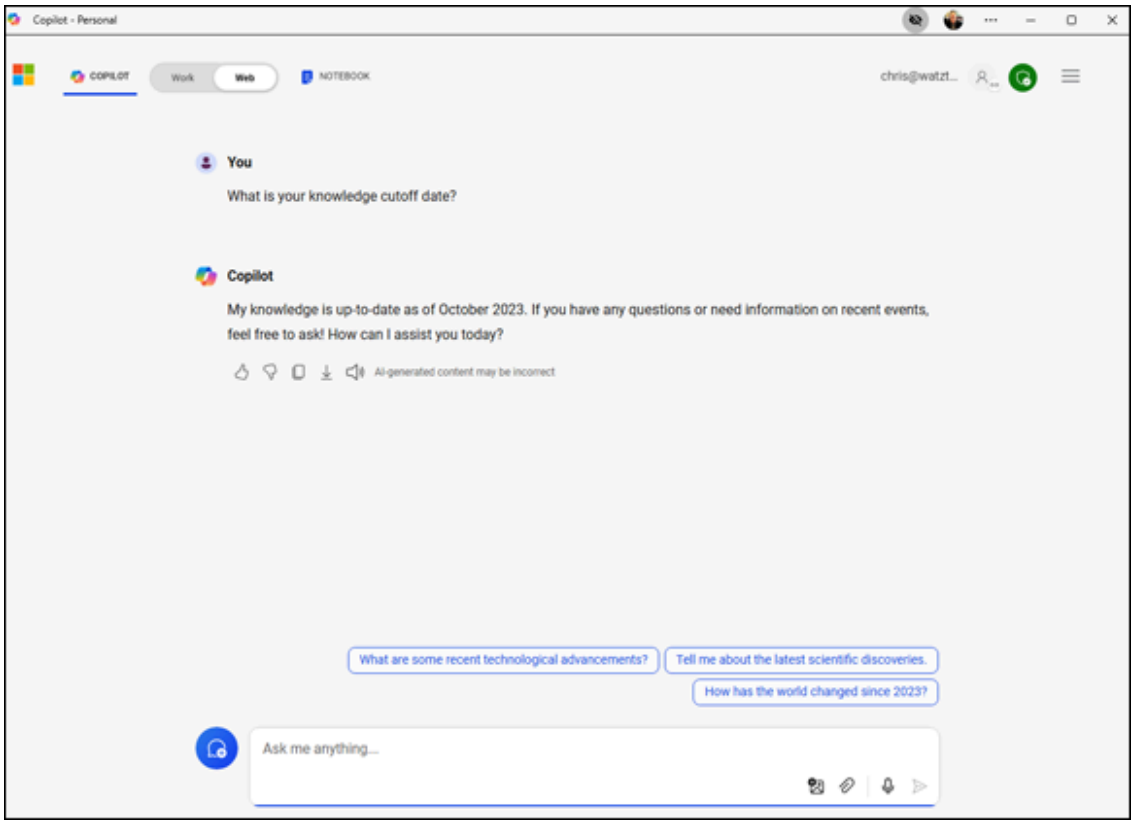


FIGURE 5-1: Asking Copilot for its cutoff date.

Enhancing functionality with plugins

If you don't enable the Search plugin (in the personal version of Copilot) or the Web Content plugin (in the work/education version), Copilot won't be able to answer questions about events that have happened after its knowledge cutoff date. With the Web Content or Search plugin enabled, however, Copilot can perform Bing searches to find up-to-date information.

When I asked Copilot 365 (without the Web Content plugin) to tell me who won the Super Bowl in 2024, it couldn't tell me. When I asked who won the Super Bowl in 2022, it was able to respond (and it was correct).

With the Web Content plugin enabled, Copilot was able to give me full and accurate answers, along with details, for both years.

The Search plugin is the most flexible and important of the Copilot plug-ins. Other plugins serve more specific purposes, as you learn in the section of this chapter called "Seeing Examples of Plugins."

Two types of plugins

You find two basic types of plugins: Teams Message Extensions and API plugins. Teams Message Extensions are plugins for Microsoft Teams that are only available within Teams. API plugins are the plugins that you can use in Microsoft Copilot 365 Chat. API plugins access programs outside of Copilot (using their application programming interface) to retrieve and even change data in the external program.



TECHNICAL STUFF An *application programming interface* (API) is a standardized way that programmers make the functionality of their program available to other programs. These APIs are typically well documented, and Copilot can use API documentation to figure out how to interact with the external program.

Creating a Copilot Plugin

There are several ways to create Copilot plugins. These range from methods that require some knowledge of how to write computer code to a method you've already seen — by using Copilot Studio.

In Chapter 4, you learn how to create Copilot agents. Agents are standalone copilots that have triggers, connectors, topics, and actions. Agents may or may not use Generative AI to decide how to respond to prompts.

In this section, you learn how to create a custom action that you can enable and use in any Microsoft 365 Copilot Chat.



REMEMBER You can read more about triggers, connectors, topics, and actions in Chapter 4.

You'll create your first plugin by creating what Microsoft calls a Custom Connector action. You must have a work or school Copilot account and be granted sufficient permission by the account administrators to be able to create plugins.

Follow these steps:

1. **Go to Copilot Studio in your browser (`https://copilotstudio.microsoft.com`) and log in.**
2. **Click the Library icon on the left toolbar.**
3. **Click Add an Item on the Library screen.**
4. **Click New Action in the Extend Copilot for Microsoft 365 window.**
   The popup window shown in Figure 5-2 will appear.

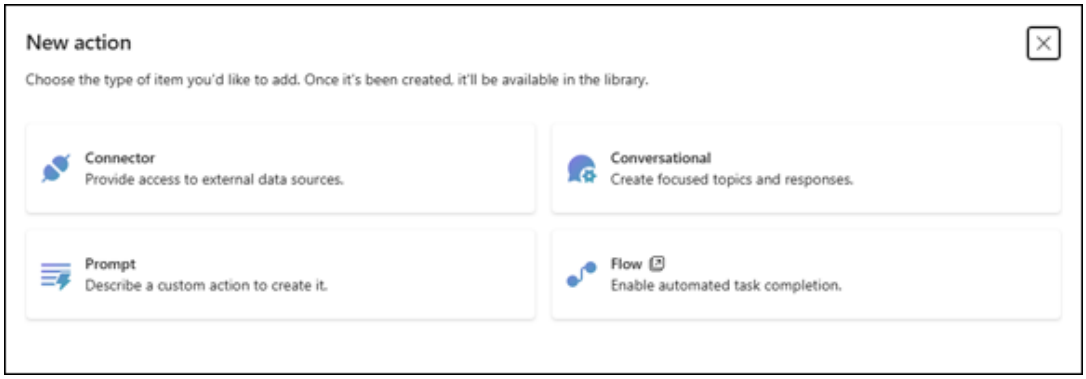


FIGURE 5-2: Select your action type.

5. **Click the Connector button in the popup window.**
   You'll see a long list of connectors. Take your time browsing through this list to see how many of the connectors are for websites and services you use. Many of the connectors require authentication and a lot of set up, but this exercise chooses one that you'll be able to use very quickly.
6. **Search for the Library of Congress connector and click on it.**
   The next screen, shown in Figure 5-3, shows a description of the custom connector and asks you to select a solution. A solution is a group of actions. For example, Copilot in Excel is a solution that includes actions for things like creating rows, deleting rows, summarizing data, and so forth.
7. **For the Action Name, enter** Search the LOC for Books. **You can leave the description as it is for now.**
   The Action Name and Description are what Copilot uses to decide whether to trigger the plugin, so you should pay extra attention to these. You may need to tweak them later after you've tested your plugin.

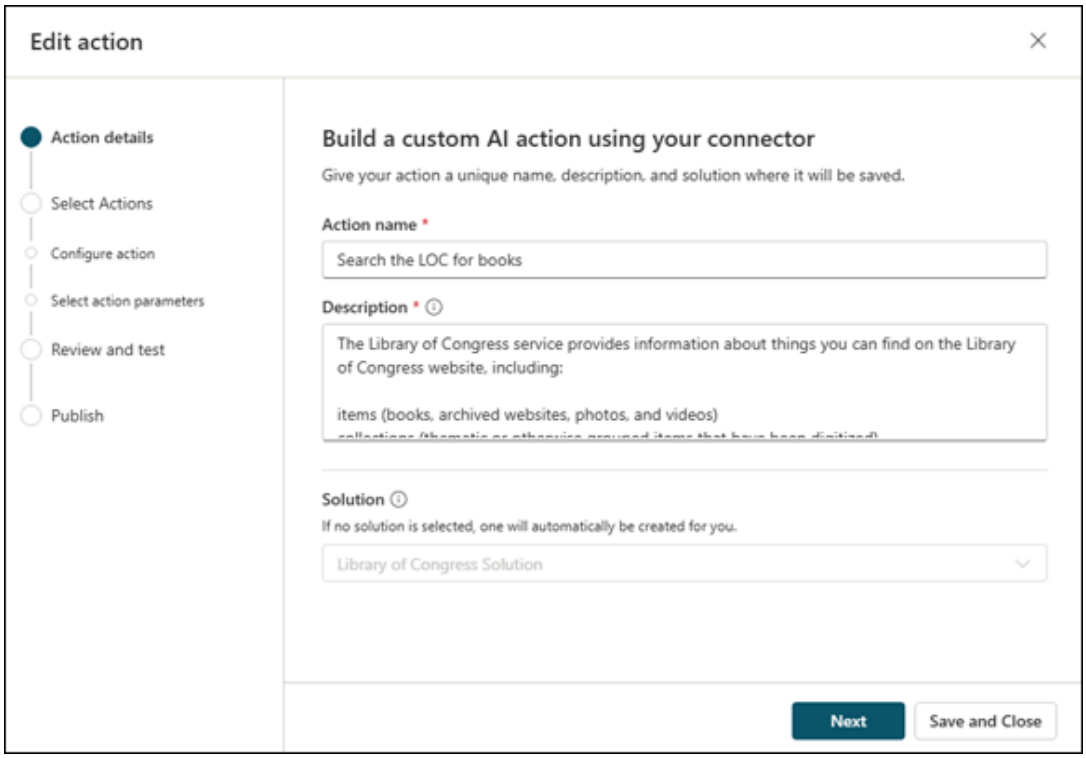


FIGURE 5-3: The connector's description.

8. **Still on the Action Details screen, leave the Solution blank, since this is the first action you're creating using the Library of Congress connector.**
9. **Click the Next button.**
10. **Choose the action named Search by Format for Items on the next screen, then click Next.**
11. **The next screen shows the inputs and outputs for the action. You don't need to change anything on this screen. Just click Next.**

    Your new custom action will be saved (which may take a minute or so), and you may need to click the Next button again when it finishes saving.

    If you now see the screen shown in Figure 5-4, you've successfully created a connector action! There are just a couple more steps before you publish it as a plugin.
12. **Click the New Connection link.**

    A connection will make the action available in Microsoft 365 Copilot.
13. **Click the Create button on the preview window that pops up.**
14. **Select the new connection you created from the drop-down menu then click the Test Action button.**

    Copilot Studio will get the plugin ready to be tested in Copilot. When it's ready, you'll see a link named Open to Test.

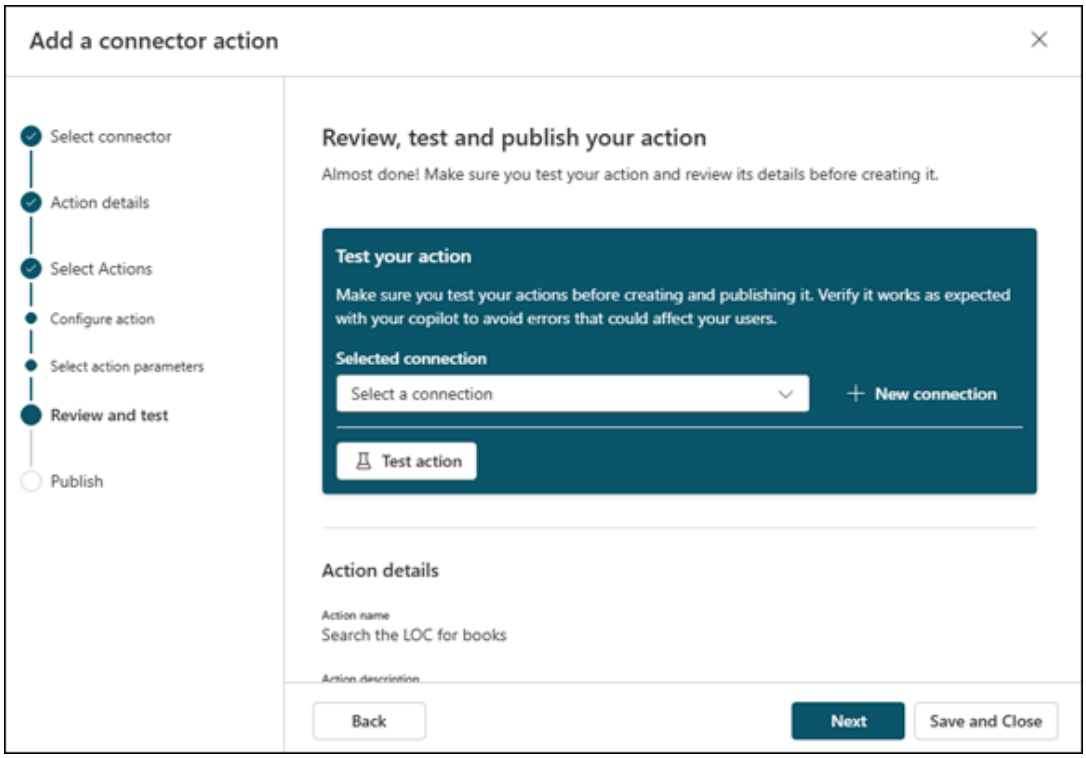


FIGURE 5-4: Your first action is saved and almost ready to test.

15. **Click the Open to Test link.**
    Copilot will launch Copilot.
16. **Click the Manage Copilot Extensions (or Plugins) in Copilot and look for your test plugin.**
    Its name will start with "Test," as shown in Figure 5-5.
17. **Make sure the test plugin is enabled, and then test your custom plugin by entering something like** Search the Library of Congress for items about cheese.



TIP In my experiments with this plugin, asking for "items" in the prompt seemed to work better than asking for "books." This may be due to the words that are used in the connector's description.

If everything works, you can return to Copilot Studio and click the Next button to publish your connector action.



TIP Many factors influence whether a Copilot plugin works as you expect it to. If you don't get the results you're expecting, try changing the plugin's description, try out a different action from the same plugin to see if that one works, change the prompt you're testing with, or just wait an hour and try again to see if the problem is outside of your control.



FIGURE 5-5: Your test plugin is now installed.

When you enable the connector action, it will be available to you whenever you use Microsoft 365 Copilot in Teams. You can also share your plugin with other people in your organization.



Want to make your plugin available to anyone in the world? At the time of this writing, there isn't a process in Copilot Studio for doing that. However, it may be possible in the future. Because of the possibility of security or privacy concerns, publishing plugins for the world to find and install will likely require additional testing and approval.

Seeing Examples of Plugins

In the personal version of Microsoft Copilot 365, you can select up to three plugins to enable at a time from the list of plugins that appears when you open the Plugins sidebar, as shown in Figure 5-6.

In the work or educational version of Copilot, the account administrator can enable additional plugins using the integrated apps settings in the Microsoft 365 Admin Center, as shown in Figure 5-7, or you can create plugins for your own use, as you read about in this chapter.

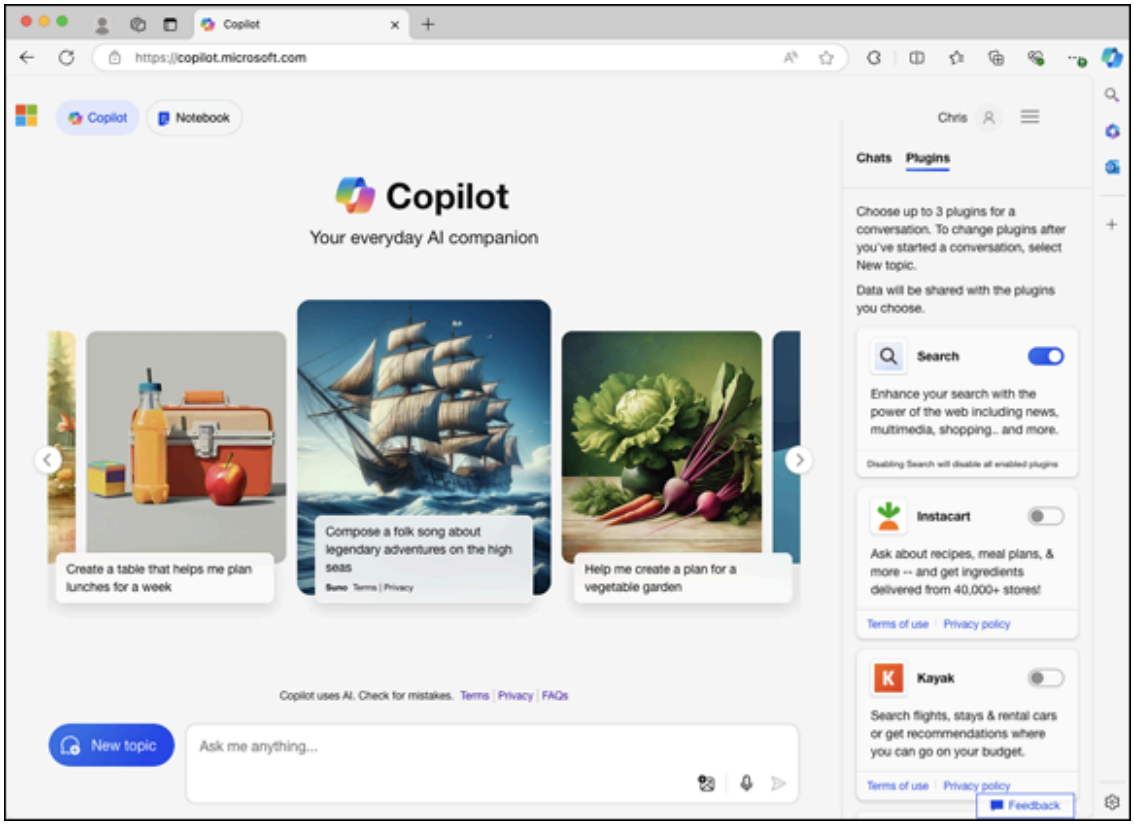


FIGURE 5-6: The Plugins sidebar.

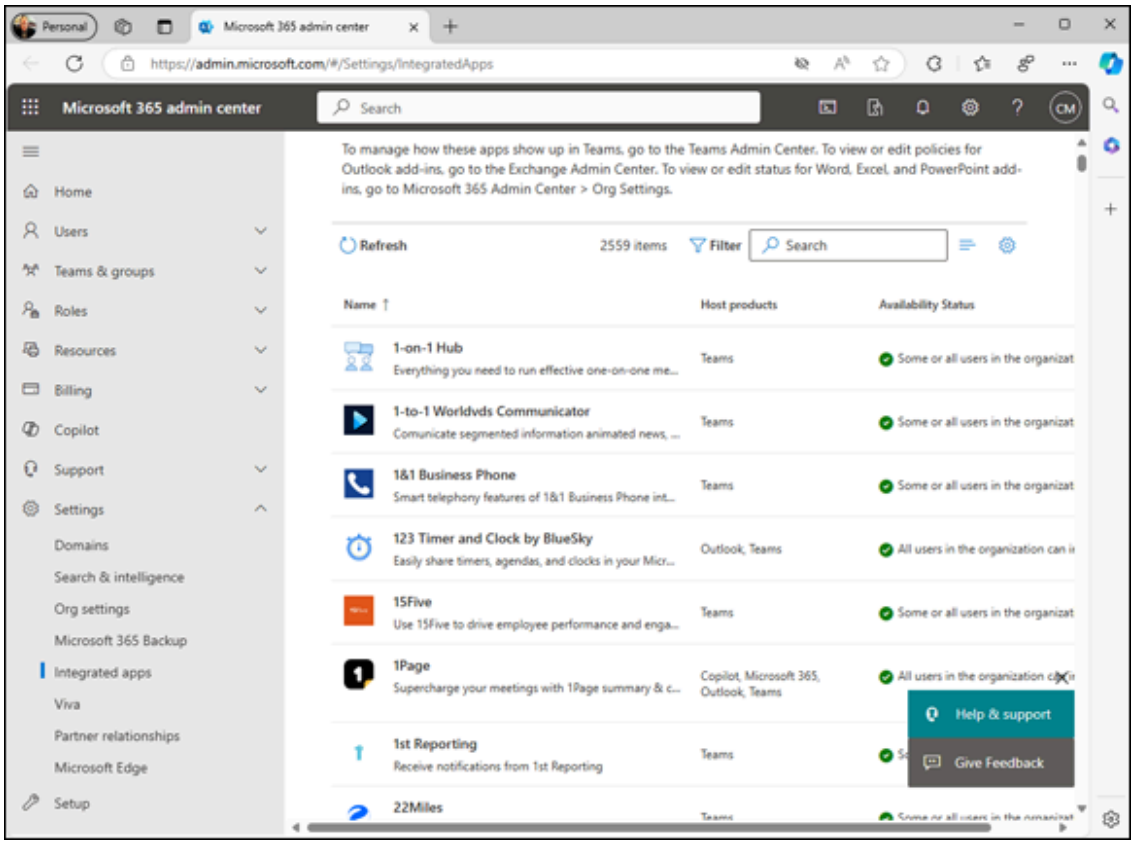


FIGURE 5-7: The integrated apps settings in Microsoft 365.

Where can plugins be used?

The big picture of plugins is still developing, and figuring out where plug-ins work and don't work can be confusing. In this section, I attempt to make a comprehensive list of the places where plugins do and do not work (at the time of this writing):

- Copilot in Microsoft Office applications (for macOS and Windows) can't use plugins.
- With the free version of Copilot (at `https://copilot.microsoft.com`) you can enable up to three plugins selected from a short list of API plugins (refer to Figure 5-6).
- Subscribers to Copilot Pro can use the same group of plugins as free users of the free version.
- Subscribers to Microsoft 365 Copilot can't use the plugins that are available to free users or Copilot Pro users.
- On Windows PCs, Microsoft 365 Copilot subscribers can enable the Web Content plugin when they're using the Work chat at `https://copilot.microsoft.com`.
- In Microsoft 365 apps, subscribers to Microsoft 365 Copilot can enable the Web Content plugin, which gives Copilot access to Bing.

- If you create your own Copilot agents, which you learn about in Chapter 4 of this book, those agents can't use plugins.
- Microsoft 365 Copilot Chat in Teams has access to custom agents created by people in your organization, and you can use an extensive list of extensions as well as custom plugins that you or others in your organization create.

Is your head spinning from trying to figure out where and when plugins work? Mine, too. I expect this will get smoother in the future, but for now, the best way to experiment with the widest variety of plugins is to use the free version of Copilot at `https://copilot.microsoft.com` along with a work or school subscription to Microsoft 365 Copilot Chat in Teams (aka BizChat).

Popular plugins for Copilot in Teams

Message extensions in Microsoft Teams are a way to allow Teams users to interact with other apps by clicking a button or typing in the Teams search bar. With a Microsoft 365 Copilot subscription, message extensions can be used as Copilot plugins.

When Copilot is enabled in Teams, not only can human users of Teams use these enabled message extensions, but Copilot can, too!

The possibilities for things that can be done with this capability are enormous. In this section, you learn about some of the more popular Copilot extensions that you can enable in Teams. But I only have space to touch on a few out of the hundreds of message extensions that have been created.



TIP You can explore available Copilot extensions in Teams by clicking the Manage Extensions icon in the prompt input area of the Copilot chat in Teams.

In the menu that opens when you click the Manage Extensions button, you'll see a button labeled Get Copilot Extensions, as shown in Figure 5-8.

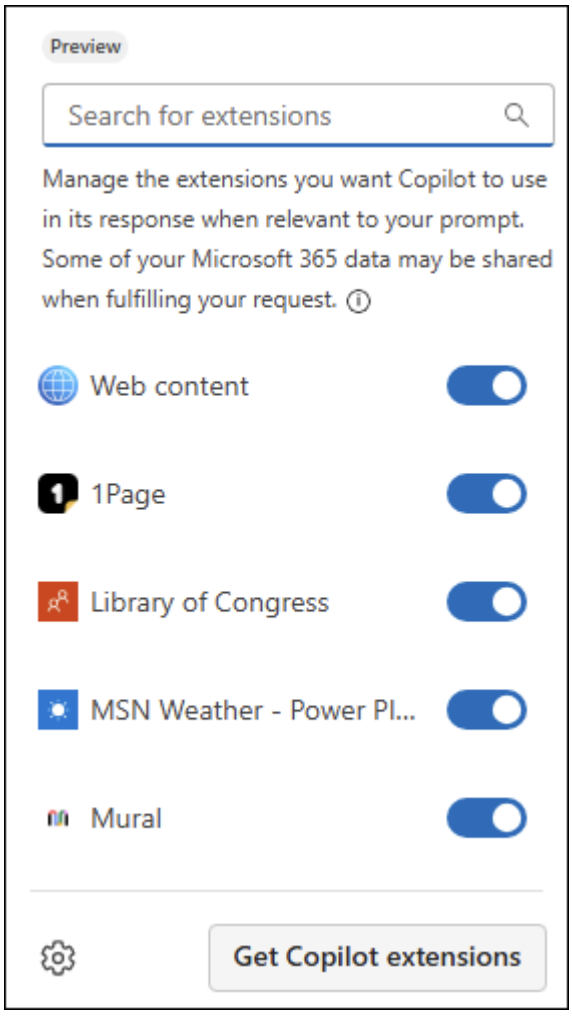


FIGURE 5-8: Get Copilot extensions.

Clicking this button opens the Teams app store, where you can explore all the available apps for Teams. Since we're only exploring Copilot Extensions at this point, I recommend you click the Filter link in the upper-right corner of the screen and narrow down the choices, as shown in Figure 5-9.

A few of the more popular Copilot Extensions in Teams include

- **Mural.** Mural is a visual collaboration tool that allows teams to work together in a digital whiteboard environment. The Mural Copilot Extension allows you to ask Copilot questions about projects you're working on in Mural, ask for template recommendations, and search your projects.

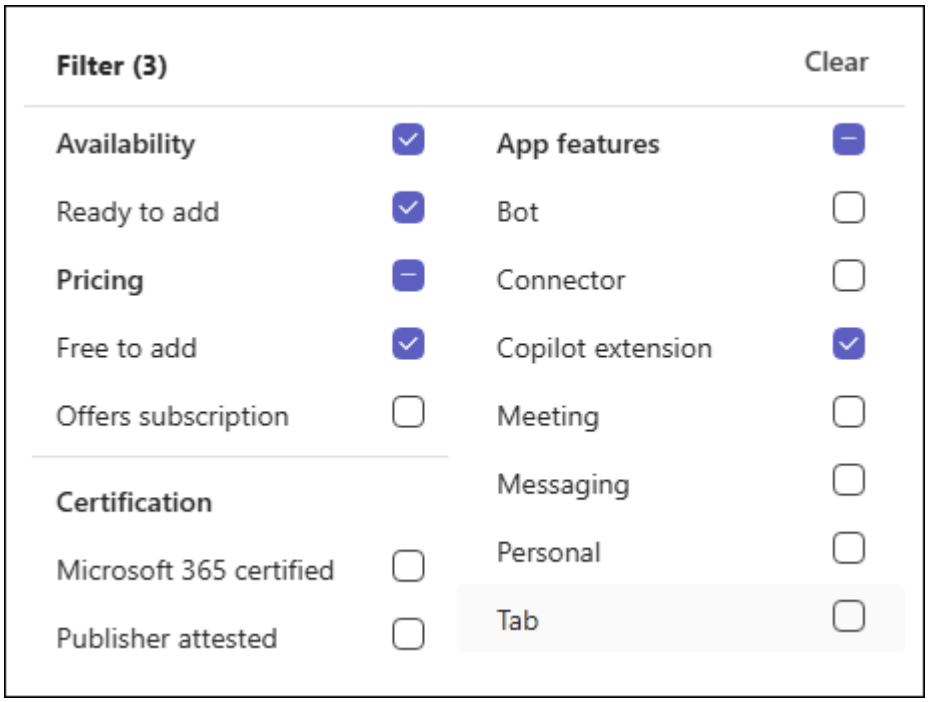


FIGURE 5-9: Filtering the apps.

- **Dropbox.** Dropbox is a popular cloud file storage and sharing app. With the Dropbox extension enabled, you can share, upload, search, and preview files in Dropbox from within Teams and ask Copilot about files you've stored in Dropbox.
- **Monday.com.** Monday.com is a project management tool. When you install the `Monday.com` extension, you gain the ability to use many of `Monday.com`'s features from within Teams and to ask Copilot questions about your projects.

Considering the Future of Plugins

Plugins give Copilot access to data outside of the data it was trained on. The idea is simple, but the potential for changing how you work is impressive. Of course, there's always a downside, and plugins can be fickle and unpredictable. The key to using plugins at this early stage of development is to be patient, understand what they can and can't do, and know when the job you want to do would be better accomplished using a traditional tool such as a search engine.